

VENKATA GANJI

AI INGESTION, INFERENCE & TOOLING SRE
DISTRIBUTED SYSTEMS SME | MISSION-CRITICAL MIGRATIONS



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Architecture Portfolio | offroadguy.github.io

AI ingestion, inference, and tooling SRE and distributed systems SME with 12+ years of production engineering experience. Owns platform design, automation, performance, and recovery across GPU-backed LLM serving, vector retrieval, and petabyte-scale data systems. Leads mission-critical distributed systems migrations and turns complex production problems into measurable improvements in reliability, latency, and cost.

ENGINEERING HIGHLIGHTS

LLM model serving vLLM, Helm/Argo CD, GPU-backed EKS	Distributed systems SME Petabyte-scale storage, ingestion and recovery
Vector database ownership 250+ nodes across Solr, Milvus, OpenSearch	Mission-critical migrations Hadoop, Solr, HBase and cloud platforms
Observability: logs to alerts Logging, metrics, dashboards, alerting	End-to-end IaC tooling Terraform, CloudFormation, Ansible, Python
Disaster recovery ~15-minute RPO / ~2-hour RTO	Performance tuning JVM/GC, query/index, cache and OS tuning
Cybersecurity telemetry PCAP, NetFlow, application logs, firewall events	

PROFESSIONAL EXPERIENCE



Apple

Apr 2024 - Aug 2026

GenAI Platform Engineering & Cloud Architect

Contract via Mphasis Corp | GenAI retrieval, model serving, ingestion, and reliability

- **LLM platform ownership.** Developed and co-owned Helm/Argo CD tooling for onboarding and deploying vLLM models on a dedicated GPU-backed EKS cluster, integrated with the ingestion and inference platform on a separate EKS cluster.
- **Model artifact lifecycle.** Implemented S3-to-FSx for Lustre staging, checksum generation and startup verification, shared model volumes, read-only serving mounts, and Kubernetes credential integration.
- **Inference serving.** Configured tensor parallelism, precision, context limits, GPU memory budgets, and LoRA adapter serving; maintained custom vLLM images and a 122B FP8 deployment configuration with four two-GPU replicas.
- **Distributed retrieval at scale.** Expanded and automated infrastructure from approximately 10 to 250+ nodes, supporting about 200 TB of vector indices across Solr, Milvus, and OpenSearch.

PROFESSIONAL EXPERIENCE | CONTINUED

**Apple | Continued**

Apr 2024 - Aug 2026

GenAI Platform Engineering & Cloud Architect

Contract via Mphasis Corp | GenAI retrieval, model serving, ingestion, and reliability

- **Performance engineering.** Reduced vector-query latency from approximately 7.0s to 1.9s through JVM/GC, OS, cache, query, and index tuning against production workloads.
- **Capacity and cost.** Architected a 200-node Solr migration to AWS Graviton/ARM64, maintaining query throughput and approximately 3,200 vCPU / 12.8 TB RAM capacity while reducing cost 19.8% (\$266K+ annualized).
- **Production reliability.** Helped reduce incidents from approximately 20/day to 2/week through observability, automation, and recurring-failure remediation; served as escalation SME for pod crashes, memory pressure, storage failures, and latency.
- **End-to-end infrastructure ownership.** Authored and owned Solr IaC and lifecycle tooling with Terraform, CloudFormation, Ansible and Python. Productionized 10+ GenAI microservices across Dev/UAT/Production, including manifests, routing, observability and deployment diagnosis.
- **Platform efficiency.** Designed and productionized an EC2 Mac document-conversion platform with application developers, saving approximately \$16K/month.

CLOUDWICK TECHNOLOGIES INC | APR 2014 - APR 2024

Full-time employer | Selected client engagements and engineering leadership

**Amgen**

Sep 2023 - Apr 2024

AWS Cloud Solutions & Migration Engineer

Cloudwick client engagement | Cloud modernization and GenAI platform infrastructure

- Migrated 100+ critical AWS accounts from legacy Landing Zone patterns to Control Tower and Organizations; built repeatable Python, CloudFormation, CDK, Service Catalog, and SSM migration workflows.
- Implemented secure S3 data-migration workflows with boto3, S3 Batch Operations, IAM, and KMS; coordinated account owners and business stakeholders through controlled production changes.
- Built and performance-tuned vector-database infrastructure, resolved GenAI ingestion-pipeline failures, and developed Kubernetes deployment pipelines and Python monitoring integrations.

CLOUD ARCHITECTURE & DISTRIBUTED DATA | CLOUDWICK CONTINUED

FreddieMac

Freddie Mac

Jan 2019 - Aug 2023

AWS Cloud Solutions Architect & Infrastructure Consultant

Cloudwick client engagement | 5+ projects across 3 teams; technical leadership for 3-8 engineers

- **Technical leadership.** Led three mission-critical cloud migrations across distributed data and live OLTP/web platforms, coordinating architecture, implementation, data movement, performance testing, and production transition.
- **Distributed data migration.** Designed the transition of a mortgage-document platform from Cloudera Hadoop/Solr to Elasticsearch/OpenSearch and S3; led Spark migration-job development for Solr-to-Elasticsearch and HBase-to-S3 data movement.
- **Ingestion architecture.** Designed Databricks, AWS Glue, Talend, and S3 ingestion/data-warehouse frameworks for upstream and downstream systems; served as senior escalation SME for platform operations and release failures.
- **Recovery engineering.** Delivered a production hybrid-cloud DR environment within six months, with approximately 15-minute RPO and 2-hour RTO using cross-region replication, S3, Airflow, Python, and Elasticsearch.
- **Platform automation.** Developed reusable Terraform, CloudFormation, and Ansible provisioning pipelines; built Python/Airflow automation for snapshots, migration, and instance bootstrapping, and partnered on Dynatrace performance testing at scale.
- **Application modernization and security.** Designed EKS deployment architecture for a legacy Oracle/WebLogic application and federated authentication across four environments; partnered with InfoSec on secure on-premises-to-S3 connectivity.

TECHNICAL EXPERTISE

AI serving and orchestration: vLLM, LoRA adapter serving, GPU resource configuration, Helm, Argo CD, Kubernetes/EKS, Kustomize, Docker

Distributed data and performance: Solr, OpenSearch/Elasticsearch, Milvus, Hadoop/HDFS, HBase, Kafka, Spark; JVM/GC, query/index tuning, capacity planning

Automation and reliability: Python, Bash, Terraform, CloudFormation, Ansible; AWS, Prometheus, Grafana, Splunk; incident response, observability, HA/DR

DISTRIBUTED SYSTEMS & TELEMETRY | CLOUDWICK CONTINUED**Cyber Data Lake / Amorphic**

Nov 2016 - Dec 2018

Lead Solutions Engineer

Telemetry ingestion, distributed analytics, and performance engineering

- Designed a three-rack test bed with IDS/IPS and packet brokers for PCAP and NetFlow capture; deployed and performance-tuned 40 Gb/s packet-capture technology across hardware and software boundaries.
- Implemented telemetry ingestion spanning PCAP, NetFlow, application logs, and firewall events to build a correlated cybersecurity data lake.
- Integrated Corelight and IXIA telemetry with the cybersecurity data platform; operated Hadoop, Kafka, Spark, Solr, HBase, and Elasticsearch components supporting ingestion and distributed analytics.
- Built Prometheus monitoring for packet-collector agents on endpoint servers; served as operations and performance SME for Databricks/EMR, ELK, and standalone Hadoop.

SELECTED CLIENT ENGAGEMENTS | 2016**SimplyMeasured | Senior Operations Solutions Engineer | Oct 2016**

Benchmarked and tuned an AWS big-data platform and authored scale, reliability, and DR guidance.

Daimler AG | Senior Operations Solutions Engineer | Jun - Oct 2016

Built and tuned two enterprise Hadoop platforms; automated deployment with Ansible and implemented Kerberos, Ranger, Knox, and SSL security.

Classmates / The Battle Group | Senior Operations Solutions Engineer | Feb - May 2016

Delivered Hadoop production-readiness reviews, benchmarking, security, automated builds, and operational handbooks; built a custom Spark Livy job-server solution for Classmates.

**Amobee / Turn**

Apr 2015 - Jan 2016

Operations Solutions Engineer - Hadoop / HBase

- Operated and expanded Hadoop environments with up to approximately 400 nodes per production cluster and 30 PB total capacity, including 24/7 support, upgrades, and disaster recovery.
- Executed a major refresh/expansion involving approximately 400 node additions/removals; implemented NameNode and JobTracker HA, Oozie migration/DR, and disk-failure checks.
- Built Python tooling on the Cloudera Manager API for NOC service management; implemented JMX telemetry with OpenTSDB/Grafana and automated deployment and performance-management tooling with Puppet.

**Limelight Networks**

Apr 2014 - Apr 2015

Operations Engineer - Cloud / Big Data

- Designed a Hadoop billing platform with HA/DR and maintained an approximately 80-node video analytics cluster through upgrades and production troubleshooting.
- Tuned Hive and MapReduce configurations against performance benchmarks; built Python/JMX monitoring, metadata backups, and Oozie data-processing workflows.

EDUCATION

Bachelor of Technology, 2011.

Graduate coursework in Computer Science: Texas A&M University, 15 credits (2014); Montana State University, 15 credits (2013).